Harava

How to collect with Fieldwatch and read what your phone heard: import, map, devices, layers and the classroom phone check

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harava is Finnish for *rake* · on your own computer or at `harava.oh2gax.org`

In short. The Fieldwatch app on an Android phone listens to the Wi-Fi access points and Bluetooth LE gadgets around you: routers, phones, earbuds, cars, trackers such as AirTags, smart glasses, sensors. It only listens; it never connects to anything. Harava takes what Fieldwatch wrote down and lets you look through it on your computer:

- **what was around you and where** — the map with your path and one pin per radio, a timeline of the day, and a detail card for every radio (Section 8);
- **what kind of radios they were** — device layers such as Glasses & VR, Trackers, Vehicles and Phones, each with its own switch (Section 9);
- **every radio in one list** — the Devices page with filters, tags and CSV export, and *My devices*, which hides your own router, PC and phone (Sections 11–13);
- **are all phones in the box?** — the Phone check for a classroom (Section 14).

Section 2 tells how to collect with Fieldwatch so that nothing is missed (use *both* the log and a Sit). Section 16 explains what the data can and cannot tell, and Section 19 is a one-page quick reference. On a phone, open `/app` (for example `harava.oh2gax.org/app`): the *Harava Mobile app* shows the same data one view at a time and uploads exports straight from the phone; it has its own guide under User guides.

About the figures. All screenshots are from Harava with the author's own Fieldwatch log of 8 November 2026: a day at home with two short car trips. They were taken in the light theme, with Mask addresses on (Section 15: the second half of every address and all Wi-Fi names are hidden) and *without the map background*, so the map shows only the path and the pins on a plain grey surface. On your screen the map has streets under them. The numbers on your screen will be different.

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1 Fieldwatch and Harava in a nutshell

1.1 What Fieldwatch hears

Every Wi-Fi access point sends a short *beacon* about ten times a second, with its address (BSSID) and usually its network name (SSID). Every Bluetooth LE gadget that wants to be found sends *advertisements*: its address, sometimes a name, a company number and a few bytes of data. Fieldwatch on your phone picks these up and writes down, for each one, *what* was heard (address, name, signal strength, the raw bytes) and *where your phone was* at that moment (GPS).

Signatures. Fieldwatch compares every radio with its catalog of about 260 *signatures*: patterns in the address, the name, the company number or the data that point to a kind of device — *Apple AirTags*, *Ray-Ban / Meta glasses*, *Tesla*, *Flock Safety Cameras*, *Ruuvi*, and so on. A signature is a pattern match, not an identity: it says "this looks like an AirTag", not whose AirTag it is.

Random addresses. Phones, PCs and earbuds change their Bluetooth address every few minutes to protect their owners' privacy. One phone can therefore appear as many radios during a day. Such addresses carry the flag `RAND`.

1.2 The log and the Sit

Fieldwatch keeps two different records, and Harava reads both:

	Log (Reports → Log export)	**Sit** (Reports → Start sit … End sit, then Sit export)
What	hear by hear: time, signal, raw data, vendor, position	one row per radio for the whole sit: first and last heard, number of hits, weakest and strongest signal, signatures, *Mine*
Gaps	a row the first time a radio is heard, then every 25th hear, **at most 16 rows per scan**; a radio heard briefly in a crowded place can be missed	keeps **every** radio, up to 6000
Needs	Write to disk on	nothing extra
Gives Harava	the timeline, the path, the signal chart, the Phone check	the complete list, and what the log missed

Table 1. The two records of Fieldwatch. Use both.



Figure 1. From the air to the screen. Fieldwatch listens and exports; you copy the files to your computer; Harava imports them into its own database and shows them.

Words used in this guide. **Radio**: one Wi-Fi access point or Bluetooth gadget, as one address. **Hear**: one row of the log (one moment when Fieldwatch wrote the radio down). **Hear-point**: where *your phone* was at a hear; the place where a radio was heard loudest is its pin on the map. **RSSI**: received signal strength in dBm; −30 is very close, −60 in the same room, −90 and weaker far away or behind walls. **Signature**: a Fieldwatch pattern for a kind of device. **Layer**: one kind of device on the map (Section 9). **Separated**: a tracker that is away from its owner's phone.

2 Collecting data with Fieldwatch

Before you go. In Fieldwatch, turn on Settings → Write to disk (the log) and Tag detections with GPS (positions). Keep the phone charged: listening and GPS use the battery, and the phone gets warm.

Start a Sit. Reports → Start sit names a session: a lesson, a walk, a drive or a whole day. While it runs, the title shows FIELDWATCH · SIT. End it with End sit. A sit can run all day.

Export both. Afterwards export:

1. Reports → Log export, format Log file — CSV (JSON lines also works);
2. Reports → Sit export for the sit, format Log file — CSV (or JSON lines).

Share or save the files and copy them into the `data` folder of the Harava project on your computer (sub-folders are fine, e.g. `data\kml`). KML exports can be imported too, but CSV carries more. The `data` folder is never committed to git: your exports contain real addresses and your locations.

Why both? The Fieldwatch log writes at most 16 rows per scan and writes a radio again only at every 25th hear. In a crowded place — a classroom, a shopping centre, a busy street — a gadget that is heard for only a few seconds can be left out of the log altogether. The Sit keeps every radio. With both, Harava shows the timeline from the log *and* marks the radios that only the Sit caught (sit only).

Keep the screen on for Bluetooth. When the phone's screen goes dark, many phones — Samsung above all — *park the Bluetooth scan*, while Wi-Fi scanning goes on. A Sit does not help: it is the same scan. In the author's logs of 8–9 November Bluetooth was paused for 70–80 % of the time, because the screen was switched off while Fieldwatch ran. For a walk, a drive or a Sit where Bluetooth matters, keep Fieldwatch in front with its Keep screen on setting (dim the screen, carry a power bank), and give Fieldwatch Allow background usage and Unrestricted battery. Harava shows on the timeline where Bluetooth was paused (Section 8.2).

Fieldwatch's own KML files and Google Earth. Fieldwatch version 1.1.42 writes hidden Wi-Fi names as invisible zero characters into its KML files, and Google Earth refuses to open them. Harava removes those characters when it imports, so the files load fine.

3 Installing and starting Harava

Harava is a small program in Python that shows its pages in your web browser. It runs on your own computer, where everything stays with you, or on a server such as `https://harava.oh2gax.org`, where you sign in (Section 4). Setting up a server is described in the *Harava Admin and server guide*.

1. Install Python 3.9 or newer from `https://www.python.org/` (on Windows, tick *Add python.exe to PATH*).
2. Double-click `start_harava.bat` in the Harava project folder. The first time it installs the one library it needs (Flask).
3. The browser opens at `http://127.0.0.1:5057`. Close the black window (or press Ctrl+C in it) to stop Harava.

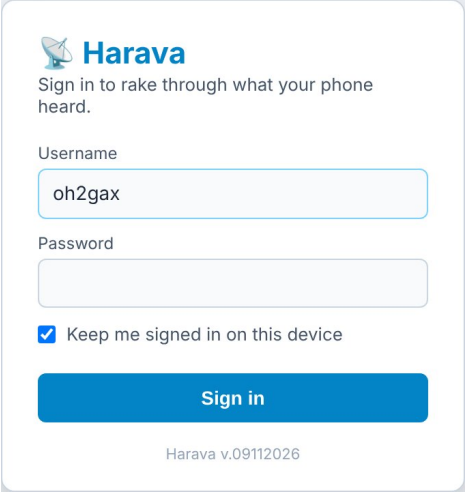
The map key. The map background comes from CARTO. Without a free key it shows an "API key required" watermark. Put the key into `api_keys.py` (copy `api_keys.py.example`) as `CARTO_API_KEY = "..."`, exactly as in WSPR Logger. The file never leaves your computer.

Settings. Copy `config.ini.example` to `config.ini` to change the port, the folder that Scan folder reads (`data`), whether a browser opens at start, and how far apart two of your stops must be before a radio heard at both counts as *travelled* (500 m).

File	What it is
`start_harava.bat`	starts Harava (Windows); `python harava.py` does the same
`harava.py`	the program
`harava.db`	your database: all imports, My devices; delete it to start from scratch
`harava_users.db`	the users and their permissions (only when users exist)
`api_keys.py`	your CARTO map key (local only)
`config.ini`	your settings (optional)
`doc/`	this guide, `whats_new.md` and `guides.json`
`data/`	your Fieldwatch exports; *Scan folder* reads them (not in git)
`tmp_files/`	your own working files (not in git)

Table 2. The files of Harava.

4 Signing in and permissions



On your own computer Harava opens at once. On a server, and on any Harava where an admin has added users, it first shows a sign-in page. Type your username and password; Keep me signed in on this device remembers you for 30 days. The menu then shows your name and permission under the title, and the power button next to the theme button signs you out. After five wrong passwords from the same place, Harava waits 15 minutes before it lets anyone try again from there.

Permission	What you can do
View only	look at everything: the map, the Devices list, the Phone check, the Identification page and the guides; export CSV
Analyze	also change the data: upload and import Fieldwatch exports, Scan folder, switch or delete imports, add or remove My devices, Reload database
Admin	also manage the users on Maintenance

Table 3. The three permissions. Buttons that need a permission you do not have are not shown.

Your first sign-in. The admin gives you a one-time password. Harava then opens *My account*, where you choose your own password (at least 12 characters). Later you find *My account* by clicking your name in the menu; changing the password there signs out your other devices.

Forgotten password. Ask the admin for a new one-time password (New password on the Maintenance page). Users, permissions and the Maintenance page are described in the *Admin and server guide*.

5 The page at a glance

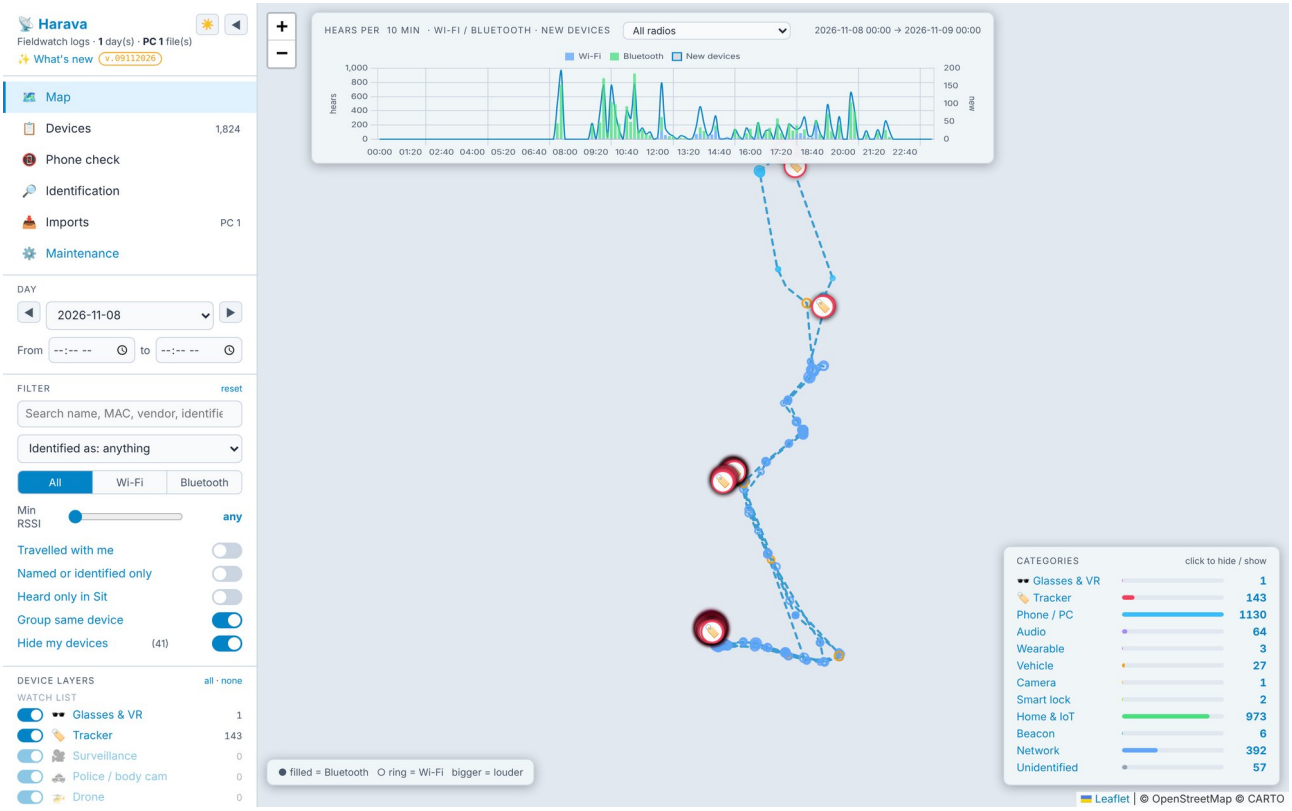


Figure 2. Harava with the log of 8 November 2026. Left: the menu. Top: the timeline of the day. Middle: your path (dashed line) and the pins of the radios. Right: the categories of the radios heard.

The menu on the left, from the top:

- **Harava** with the number of days and files, your name and permission (when you are signed in), and What's new with the version badge. The badge turns amber when there are notes you have not read. It opens one window with three tabs: What's new, User guides (this guide) and About. The sun / moon button switches between Auto, Dark and Light; the power button signs you out; ◄ hides the menu (a ► handle on the left edge brings it back).
- **Pages**: Map, Devices, Phone check, Identification, Imports, and Maintenance for admins.
- **Day** and an optional **From–to** time; ◄ ► step a day. All days shows everything.
- **Filter**: search, Wi-Fi / Bluetooth, minimum RSSI and five switches (Section 10).
- **Device layers**: one switch per kind of radio (Section 9).
- **Map layers**: the phone path, the timeline graph, the categories card, Spread stacked pins and Mask addresses (Section 15).
- **Shown now**: devices, hears, Wi-Fi, Bluetooth, trackers and travelled radios after the filters.

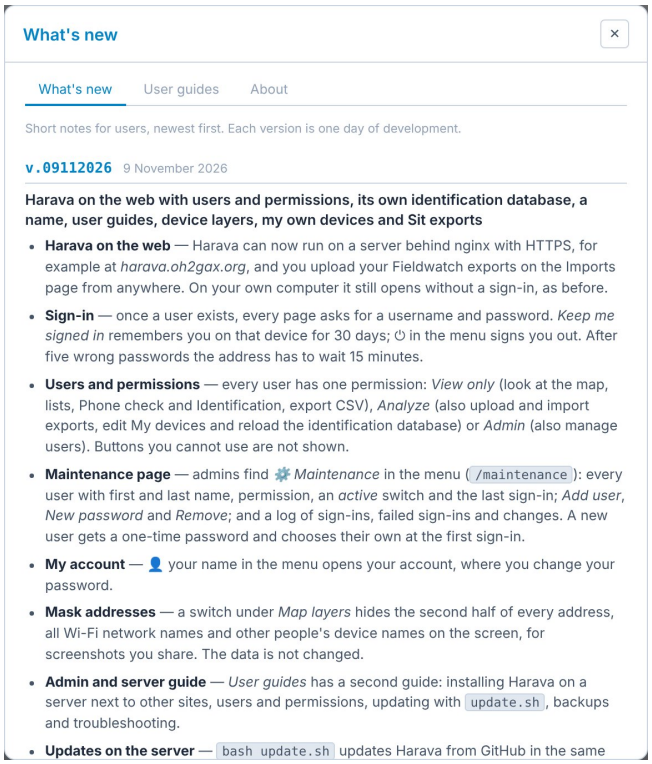


Figure 3. What's new opens one window with three tabs: the notes of each version, the user guides (PDF) and About.

6 Importing

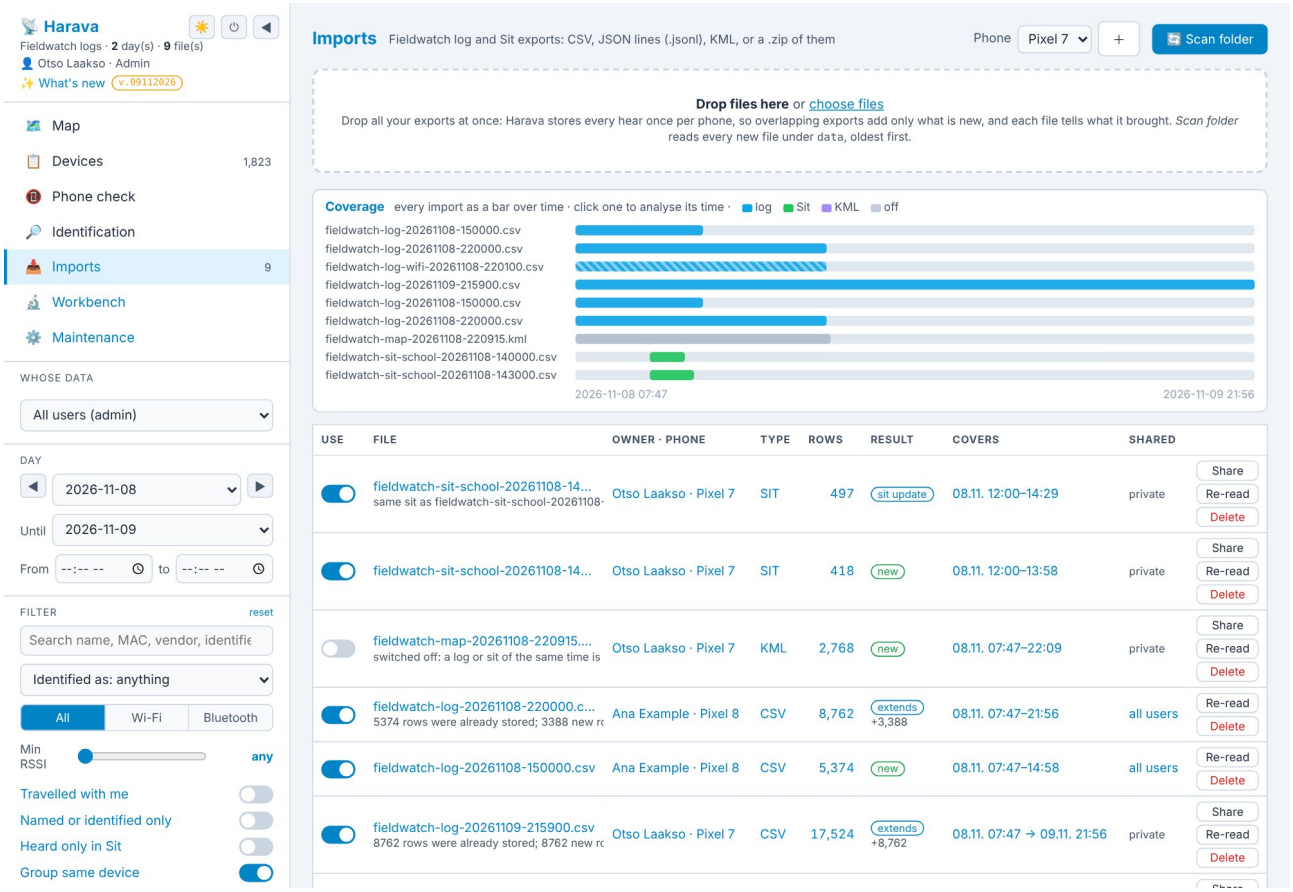


Figure 4. The Imports page seen by an admin with All users: two exports of the rolling log a day apart (*extends*), a Wi-Fi-only export (*duplicate*, striped), two exports of one running Sit (*sit update*), Fieldwatch's KML switched off, and another user's shared imports from her own phone. The coverage bars show what each file covers.

Drop everything. Drop files on the dashed box (or choose files) — one file, many files, or a `.zip` of them. On your own computer Scan folder reads every new `.csv`, `.jsonl` and `.kml` in the `data` folder, sub-folders included, oldest file first. You do not need to pick files carefully: Harava sorts out the overlaps.

Why exports overlap. Fieldwatch keeps its log on the phone in 12 rotating parts of about 1 MB, and a log export copies *everything still on the phone*. An export made on Tuesday therefore repeats most of Monday's export. Harava stores **every hear once per phone** and remembers which files contained it, so a newer export adds only its new rows.

The result of each file. The Result column (and the message after an upload) tells what a file brought:

- **new** — nothing of it was stored yet;
- **extends** — a newer export of the rolling log: the rows already stored, and how many new ones ("+3,388") from which time on;
- **overlaps** — partly known, partly new;
- **duplicate** — every row was already there, e.g. a Wi-Fi-only export or the same log as JSON lines (it merges with the CSV and keeps the longer raw data);
- **sit update** — a newer export of the same running Sit: its radios are updated (more hits, last seen later), not counted twice.

The very same file a second time is skipped, even under another name.

What Harava recognises. Fieldwatch log CSV and JSON lines, Sit export CSV and JSON lines (type SIT), and KML from both (the Sit KML also brings your walking path). A KML has one point per radio and cannot be matched row by row, so a **KML** of the same phone and time as a log or Sit is switched off by itself. A **log and a Sit** of the same time stay on together: the log gives the timeline, the Sit every radio.

Phone. Choose the phone that recorded the files in Phone before importing (+ adds one). Rows are merged per phone: two phones that hear the same radio at the same moment stay two observations, and the radio gets a **2 phones** tag.

The coverage bars show each import over time: blue a log, green a Sit, violet a KML, grey a file switched off, striped one that brought nothing new. Click a bar to analyse exactly its days.

Use, Re-read, Delete. The Use switch leaves a file out without deleting it. Re-read reads the original file again — Harava keeps a compressed copy of every upload — when a new version of Harava reads more from it. Delete removes the import; rows that another import also contains stay. Radios marked *Mine* in a Sit export are added to your My devices (Section 13).

Several days at once. Under Day, Until chooses the last day of a range (Figure 5): the map, the timeline, the lists and the counts then cover all those days, from every file and without duplicates. ◄ ► move the whole range.

7 Your data and other users' data

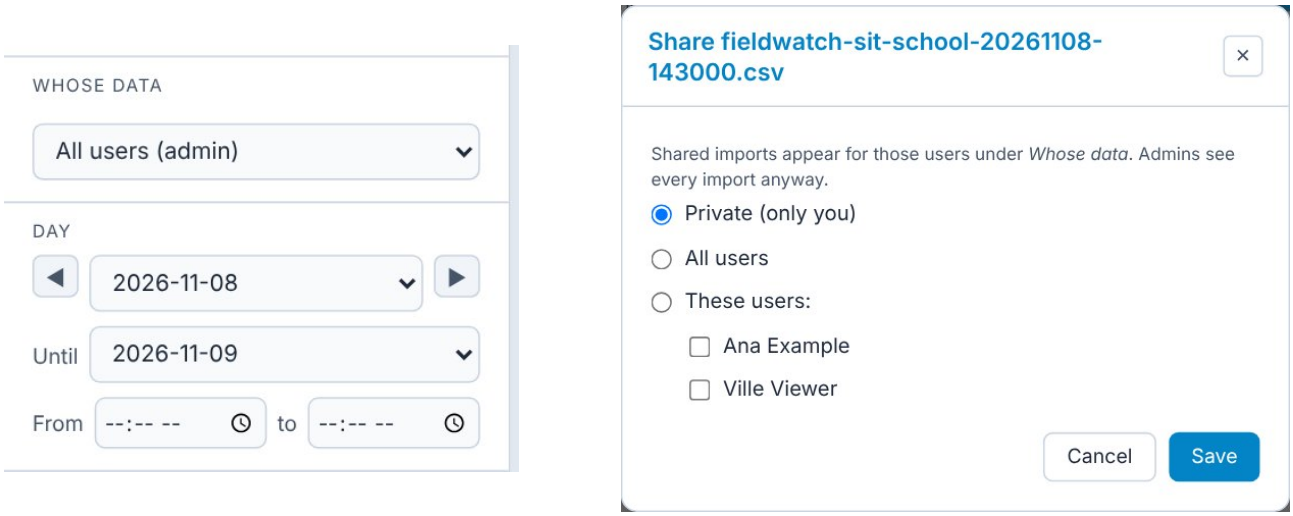


Figure 5. Whose data and a range of two days in the sidebar (left); sharing one import (right).

On a server with several users, every import belongs to the user who imported it, and **others do not see it** unless it is shared.

Whose data (in the sidebar, and in the phone app's menu) chooses what every page shows:

- **My data + shared with me** — the default: your imports and those others shared with you;
- **My data only**;
- ***Name*'s data** — one user's imports that you may see;
- **All users (admin)** — admins see everyone's data and can combine it.

Sharing. Share on your own import: *Private*, *All users*, or chosen users. A shared import appears for them under Whose data; they can look at it but not switch, re-read or delete it. Admins see every import anyway — other users' data shows where they were, so a server admin is someone the users trust.

Personal. *My devices* are personal: each user marks their own router, phone and PC, and they are hidden only from that user's views. Harava's identification database and the Workbench are shared by everyone.

8 The map

8.1 Path and pins

The **dashed line** is your path: where your phone was, in time order, from the GPS positions of the hears. A gap of more than 30 minutes without positions breaks the line.

Each radio has **one pin**, at the place where your phone heard it *loudest*. That is a good guess of where the radio was, but only a guess: the pin is always on your path, never off it.

- **filled** pin = Bluetooth, **ring** = Wi-Fi, **bigger** = louder;

- the colour is the device layer (Section 9);
- watch-list radios (glasses, trackers, surveillance, police, drones, hacking tools) get a round **icon pin** on top of the others; a tracker that is *separated* from its owner **pulses red**;
- where many radios share one spot (at home, most do), Spread stacked pins lays them out in a small sunflower so that each one can be clicked. Zoom in to see them.

Hover over a pin for its name, layer, signal and hears; click it for its details.

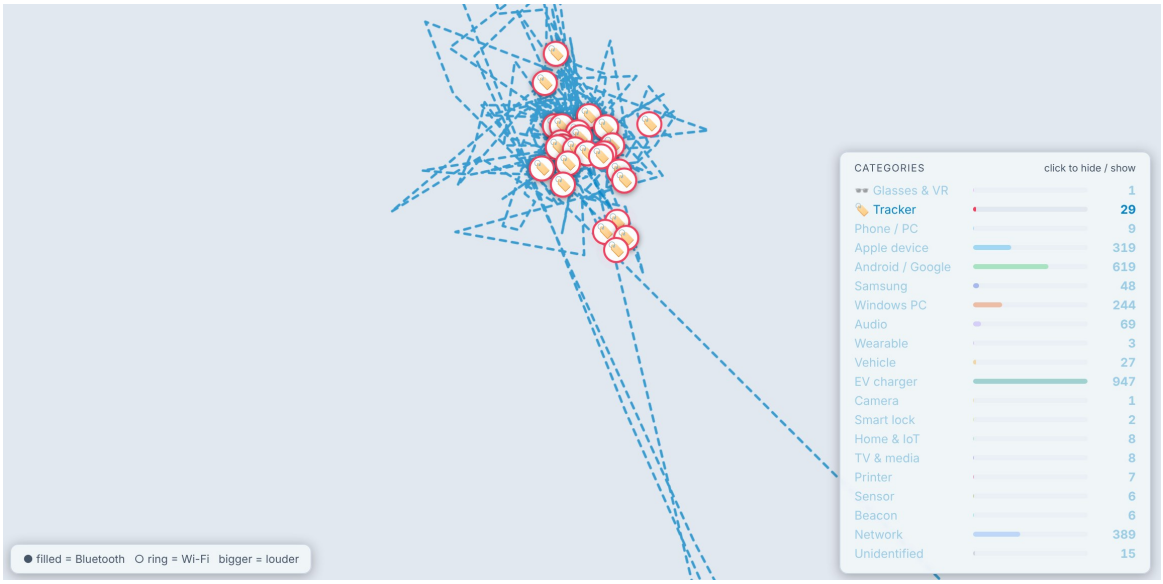


Figure 6. Only the *Tracker* layer, zoomed in at home, after Harava moved the iPhones, iPads and Macs to the Apple device layer (29 addresses left of Fieldwatch's 143). Each tag icon is one tracker address heard during the day. All were weak and heard once or twice — neighbours' tags and passers-by, nothing that followed the phone.

8.2 The timeline

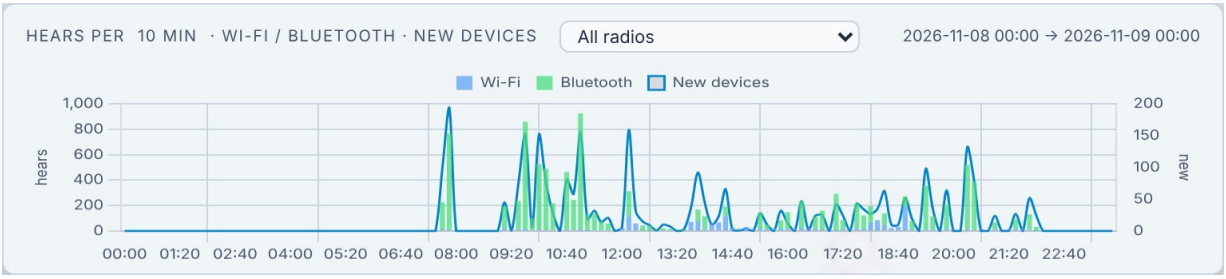


Figure 7. Hears per 10 minutes: Wi-Fi (blue) and Bluetooth (green) bars, and the number of new radios (line, right scale). The busy peaks around 08:00–11:00 are at home; the drives show as short bursts of many new radios.

The bars count the hears in each 10-minute step (each hour or day when you look at a longer period); the line counts the radios heard for the first time in that step. **Click a bar** to look at just those 10 minutes; clear the From–to times to see the whole day again.

Show one kind of device. The Show selector in the title limits the graph to one layer, for example Glasses & VR only or Tracker only; the number in brackets is how many such radios the day has. *none heard in this range* means there were none. The selector changes only the graph, not the map.

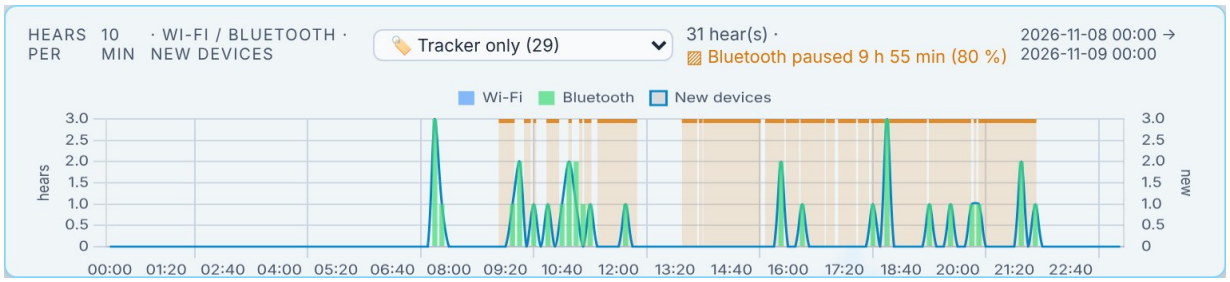


Figure 8. The same day with Tracker only: when the trackers were heard.

Bluetooth paused. An amber band over the graph marks a stretch of at least 3 minutes where the log has Wi-Fi rows but no Bluetooth rows: the phone had parked its Bluetooth scan, usually because the screen was off (Section 2). Bluetooth radios there were *not heard* — an empty stretch does not mean that nothing was there. The note in the title, ▨ Bluetooth paused …, gives the total and its share of the time the log was running; hover over it for details, and over a bar for the times of the stretch. Stretches where Wi-Fi stopped too (the phone or the log was off) are not marked. With two phones, each phone is judged on its own data. The bands always count all radios, also when Show limits the graph to one layer.

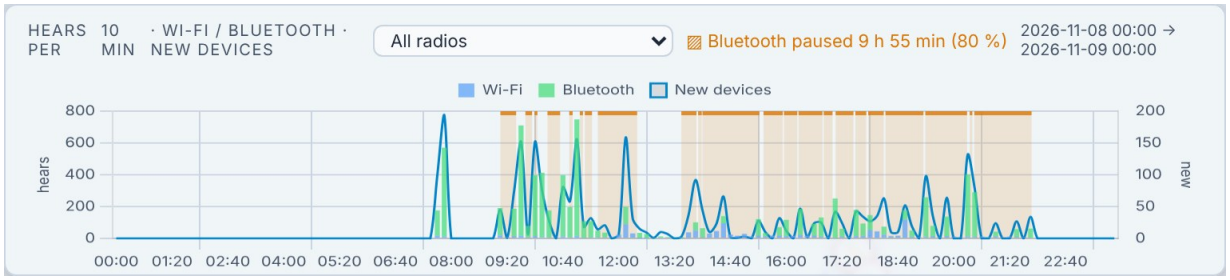


Figure 9. 8 November 2026 with the Bluetooth pauses: the screen was switched off most of the day while Fieldwatch ran, and Bluetooth was paused for 80 % of the logged time — only the narrow gaps between the bands had Bluetooth.

8.3 The selected radio



Figure 10. The D-Link 4G router selected (left), and its panel enlarged (right). Its hear-points are joined by the amber dashed line: the same address was heard at home and at stops 3.7 km away, so it *travelled* with the phone. The chart shows its signal over the day: about −70 dBmat home and −29 dBmat the first stop.

Clicking a pin (or a row of the Devices page) opens the **Selected radio** panel at the top right and draws every place where the radio was heard. Close it with ×; click any pin to open it again.

Field	What it tells
Name, address	the advertised name and the MAC address; *(hidden network)* for a Wi-Fi without a name
Pills	the layer, the radio (Wi-Fi band or Bluetooth LE), and how it travelled: *at N stops with you* or *same road*
Signatures, vendor	the Fieldwatch signatures that matched, and the maker of the address block
Tracker mode	*separated* (away from its owner) or *near owner*, read from the raw data; for Find My also the battery, in the note
Sensor	a Ruuvi tag's temperature, humidity, pressure and battery
Hears	how many times it was written down; with a Sit, also the log rows and the Sit hits
First / last	the first and last time heard (in the chosen range)
RSSI max / med	the strongest and the typical signal; *Sit RSSI* gives the sit's weakest … strongest
Places, spread	in how many places (about 200 m apart) and over what distance it was heard
Identified as	what Harava's own database says the radio is, with a confidence pill (Section 12)
Company	the company from the official Bluetooth SIG list
Same device	how many addresses the same device used, when Harava can tell
Company ID, services, raw	the Bluetooth data, for the curious
Days seen	every day in your imports on which this address appears

Table 4. The fields of the Selected radio panel.

Under the chart, + Mark as my device adds the radio to My devices (Section 13). A note tells when the address is random, when the name points to a VR headset, and when only the Sit heard the radio.

9 Device layers

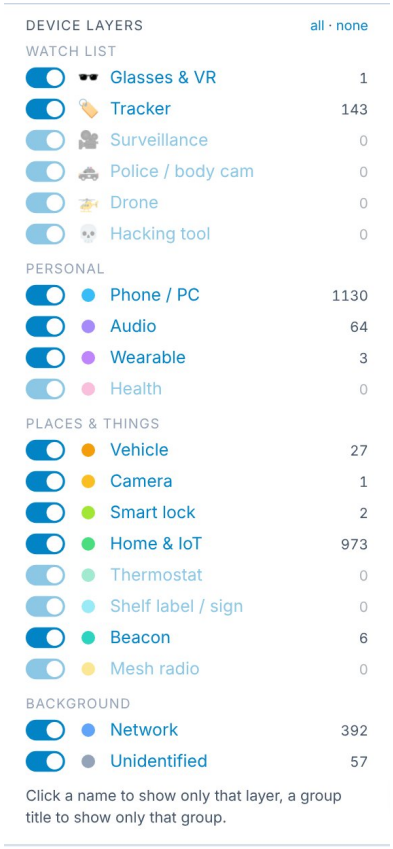


Figure 11. The Device layers section of the menu.

Every kind of device in the Fieldwatch catalog is its own layer with a switch and a count. Layers with nothing in them are dimmed.

- **Click a name** to show *only* that layer; click it again to bring all layers back.
- **Click a group title** (Watch list, Personal, . . .) to show only that group.
- all / none switch every layer at once.

Harava remembers your choice. The *Categories* card on the map shows the same counts as bars; a click on a row hides or shows that layer.

Group	Layer	Examples from the catalog
Watch list	**Glasses & VR**	Ray-Ban / Meta, Even G1, Snap Spectacles, Vuzix, RayNeo, Brilliant Frame; Meta Quest and other VR headsets
(icon pins)	**Tracker**	Apple AirTags and Find My accessories (not iPhones: see Section 16), Google Find Hub, Samsung SmartTag, Tile, Chipolo, Pebblebee, DULT
	Surveillance	Flock Safety, Axis, Dahua, traffic sensors (BlipTrack, BlueTOAD)
	Police / body cam	Axon, Digital Ally, in-car routers
	Drone	DJI, Autel, Parrot, Remote ID
	Hacking tool	Flipper Zero, Hak5 Pineapple, Pwnagotchi
Personal	**Apple device, Android / Google**	iPhones, iPads, Macs; Android phones and other Google devices, OnePlus
	Samsung, Windows PC	Galaxy phones, tablets and watches; Windows computers
	Phone / PC	what is left: phone hotspots, Fast Pair accessories
	Audio, Wearable, Health	earbuds and speakers; watches, rings, pendants; Dexcom, Omron
Places & things	**Vehicle, EV charger**	Tesla, Audi MMI, BMW, CarPlay, tyre-pressure sensors; EV charge points
	Camera, Smart lock	Arlo, GoPro, Nest; SALTO, Yale, Nuki
	Home & IoT, Thermostat	Govee, Tuya, Amazon, Hue lamps, appliances; ecobee, Nest
	TV & media, Printer, Sensor	Samsung and LG TVs, Roku, soundbars; HP, Epson, AirPrint; Ruuvi, SensorPush
	Shelf label / sign, Beacon, Mesh radio	electronic shelf labels; iBeacon, Estimote; Meshtastic, goTenna
	Two-way radio	Icom ID-52, Bluetooth speaker-microphones of Icom, Yaesu and Kenwood, PTT buttons
Background	**Network**	Wi-Fi access points and routers
	Unidentified	no signature matched: most random Bluetooth addresses

Table 5. The device layers. A radio that matches several signatures goes to the layer highest in this list (a tracker before a phone). *EV charger*, *TV & media*, *Printer* and *Sensor* are Harava's own: Fieldwatch files such radios under Home & IoT. So are *Apple device*, *Android / Google*, *Samsung* and *Windows PC*, which Fieldwatch files under one class of phones and computers.

Glasses & VR. Smart glasses and VR headsets share one layer, because Meta uses the same Bluetooth identity for its Ray-Ban glasses and its Quest headsets; Fieldwatch's *Ray-Ban / Meta glasses* signature matches both. When the name says *Quest*, *Oculus*, *Pico*, *Vive* or *Vision Pro*, Harava adds a **VR?** tag and a note. Glasses usually advertise without a name. On 8 November the only hit was "Quest 3", heard once at −96 dBm when leaving home — most likely a neighbour's headset.

10 Filters

The filters change the map, the Devices list and the counts at the same time.

- **Search** — part of a name, address, vendor, signature, layer, *Identified as* or company, e.g. `tesla`, `40:86`, `Ruuvi`, `hue`.
- **Identified as** — only one kind of identified device, grouped by layer (Section 12).
- **All / Wi-Fi / Bluetooth**.
- **Min RSSI** — only radios heard at least this loud; −70 leaves roughly what was close.
- **Travelled with me** — radios that were heard at two of your stops at least 500 m apart (tag *at N stops*), or for minutes while you moved (tag *same road*). Your own companions (watch, earbuds, power bank) show up here — and anything that followed you. See *Travelled* in Section 16.
- **Named or identified only** — radios with a name or a signature.
- **Heard only in Sit** — radios that only a Sit export has: what the log missed.
- **Group same device** — on by default; addresses that belong to one device are one row (Section 12).
- **Hide my devices** — on by default; the number is how many of your own radios are hidden now.

reset clears the filters (not the layers).

11 The Devices page

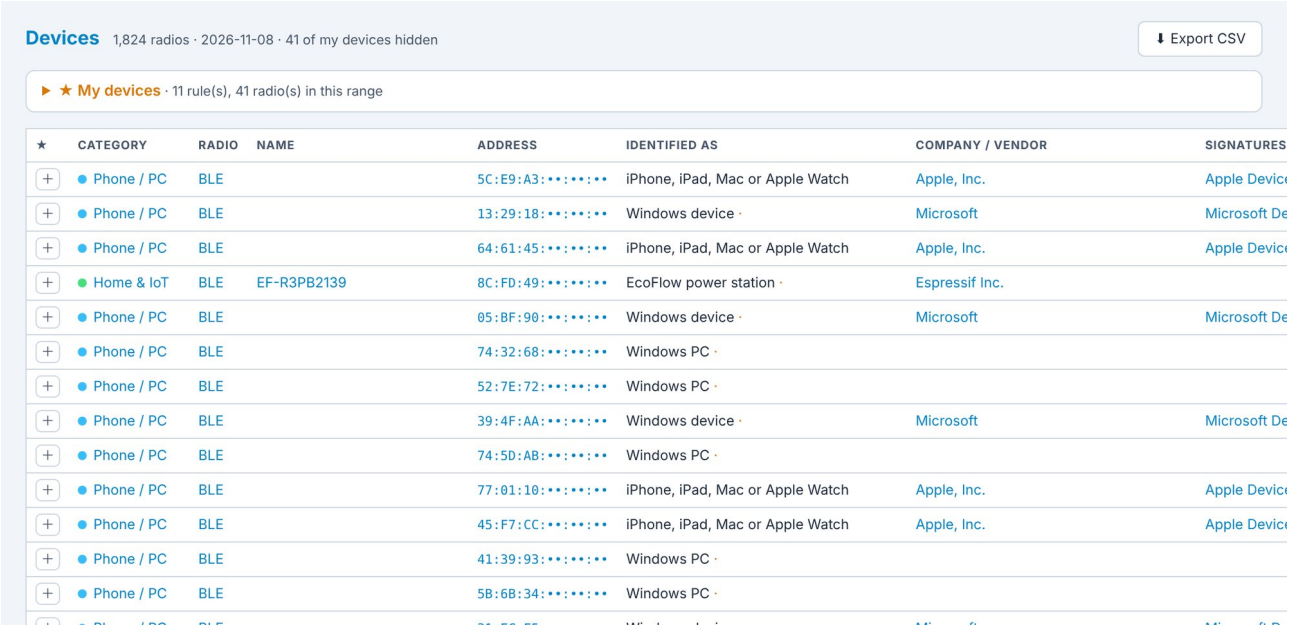


Figure 12. The Devices page, strongest first. + in the first column marks a radio as yours; the star box above it holds My devices.

One row per radio, with the same filters as the map. Click a column title to sort (again for the other way); click a row to see the radio on the map.

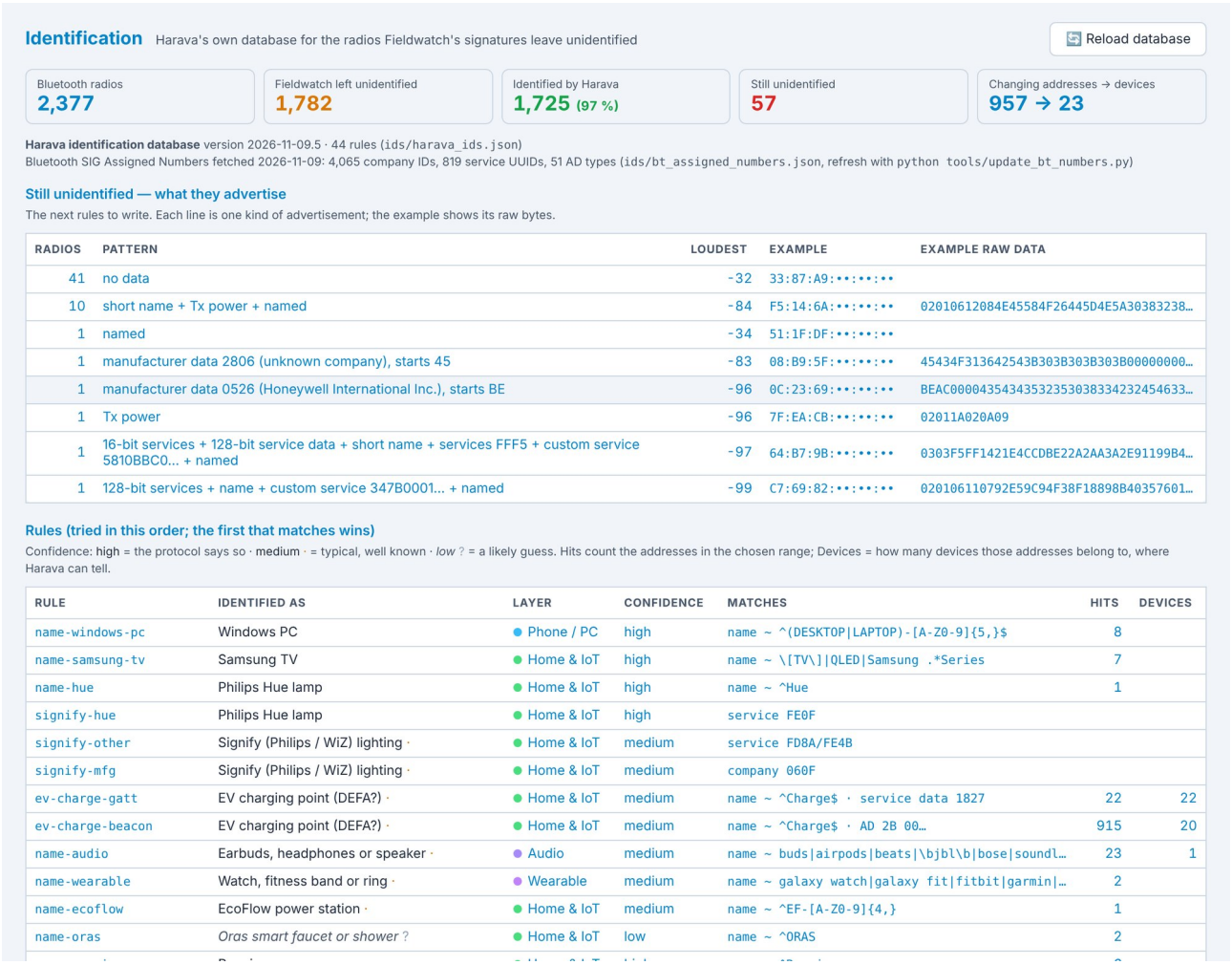
Column	Meaning
★	+ marks the radio as yours; ★ when it already is (shown when *Hide my devices* is off)
Category, Radio	the device layer; Wi-Fi or BLE
Name	the advertised name, with tags: *N addr.* (one device, many addresses), *at N stops* or *same road* (travelled), *separated* (tracker), *VR?*, *sit only*, a Ruuvi temperature
Address, Vendor	MAC address; maker of the address block (empty for random addresses)
Identified as	Harava's own identification; grey with **?** is a weak guess
Company / vendor	the Bluetooth company, or else the maker of the address block
Signatures	every Fieldwatch signature that matched
RSSI max, Median	strongest and typical signal in dBm
Hears, Ch	how often written down; Wi-Fi channel(s)
First, Last, Spread	first and last heard; distance between its hear-points
Flags	`RAND` random address, `HIDDEN` hidden network

Table 6. The columns of the Devices page.

Export CSV saves the list as it is filtered now, for a spreadsheet. The page shows the first 1500 rows; the export has all.

12 Identification

Fieldwatch's signatures name only part of what it hears. On 8 November it left 1,782 Bluetooth radios *Unidentified*. Harava has its own identification database on top of the signatures, and it named 1,725 of them (97 %).



Identification Harava's own database for the radios Fieldwatch's signatures leave unidentified

Reload database

Bluetooth radios	Fieldwatch left unidentified	Identified by Harava	Still unidentified	Changing addresses → devices
2,377	1,782	1,725 (97 %)	57	957 → 23

Harava identification database version 2026-11-09.5 · 44 rules (`ids/harava_ids.json`)
Bluetooth SIG Assigned Numbers fetched 2026-11-09: 4,065 company IDs, 819 service UUIDs, 51 AD types (`ids/bt_assigned_numbers.json`, refresh with `python tools/update_bt_numbers.py`)

Still unidentified — what they advertise

The next rules to write. Each line is one kind of advertisement; the example shows its raw bytes.

RADIOS	PATTERN	LOUDEST	EXAMPLE	EXAMPLE RAW DATA
41	no data	-32	33:87:A9:••:••:••	
10	short name + Tx power + named	-84	F5:14:6A:••:••:••	02010612084E45584F26445D4E5A30383238…
1	named	-34	51:1F:DF:••:••:••	
1	manufacturer data 2806 (unknown company), starts 45	-83	08:B9:5F:••:••:••	45434F313642543B303B303B303B00000000…
1	manufacturer data 0526 (Honeywell International Inc.), starts BE	-96	0C:23:69:••:••:••	BEAC000043543435323530383342324546 33…
1	Tx power	-96	7F:EA:CB:••:••:••	02011A020A09
1	16-bit services + 128-bit service data + short name + services FFF5 + custom service 5810BBC0… + named	-97	64:B7:9B:••:••:••	0303F5FF1421E4CCDBE22A2AA3A2E91199B4…
1	128-bit services + name + custom service 347B0001… + named	-99	C7:69:82:••:••:••	020106110792E59C94F38F18898B40357601…

Rules (tried in this order; the first that matches wins)

Confidence: high = the protocol says so · medium · = typical, well known · *low* ? = a likely guess. Hits count the addresses in the chosen range; Devices = how many devices those addresses belong to, where Harava can tell.

RULE	IDENTIFIED AS	LAYER	CONFIDENCE	MATCHES	HITS	DEVICES
name-windows-pc	Windows PC	Phone / PC	high	name ~ ^(DESKTOP|LAPTOP)-[A-Z0-9]{5,}$	8	
name-samsung-tv	Samsung TV	Home & IoT	high	name ~ \[TV\]|QLED|Samsung .*Series	7	
name-hue	Philips Hue lamp	Home & IoT	high	name ~ ^Hue	1	
signify-hue	Philips Hue lamp	Home & IoT	high	service FE0F		
signify-other	Signify (Philips / WiZ) lighting ·	Home & IoT	medium	service FD8A/FE4B		
signify-mfg	Signify (Philips / WiZ) lighting ·	Home & IoT	medium	company 060F		
ev-charge-gatt	EV charging point (DEFA?) ·	Home & IoT	medium	name ~ ^Charge$ · service data 1827	22	22
ev-charge-beacon	EV charging point (DEFA?) ·	Home & IoT	medium	name ~ ^Charge$ · AD 2B 00…	915	20
name-audio	Earbuds, headphones or speaker ·	Audio	medium	name ~ buds|airpods|beats|\bjbl\b|bose|soundl…	23	1
name-wearable	Watch, fitness band or ring ·	Wearable	medium	name ~ galaxy watch|galaxy fit|fitbit|garmin|…	2	
name-ecoflow	EcoFlow power station ·	Home & IoT	medium	name ~ ^EF-[A-Z0-9]{4,}	1	
name-oras	*Oras smart faucet or shower* ?	Home & IoT	low	name ~ ^ORAS	2	

Figure 13. The Identification page for 8 November: the counts at the top, the patterns that are still unknown, and the rules with how many radios each matched.

Keeping up with Fieldwatch. Fieldwatch gets new versions often, with new signatures and sometimes a changed log. The line *Fieldwatch signature table* at the top of the page tells which catalog of Fieldwatch Harava's layers come from. If your data has a signature name that the table does not know, an amber warning names it: Fieldwatch on the phone is then newer than Harava's table, and such radios show as *Unidentified* until the table is refreshed (`python tools/update_fieldwatch_signatures.py`, then deploy; `--check` only tells what is different).

What it reads. A Bluetooth radio advertises a few short pieces of data: a company number, service numbers, and data in a known format. Harava compares them with two files in the `ids` folder:

- `bt_assigned_numbers.json` — the official Bluetooth SIG lists: 4,065 company numbers and 819 service numbers. `python tools/update_bt_numbers.py` downloads the newest lists.
- `harava_ids.json` — Harava's own rules, for example *Bluetooth Mesh device, not set up*, *AirPods*, *iPhone / iPad / Mac*, *Samsung Galaxy phone*, *Windows PC*, *Oral-B toothbrush*. The first rule that matches wins.

A radio that Fieldwatch left *Unidentified* moves to the layer the rule gives, e.g. *Home & IoT* for a mesh light. *Identified as* and *Company / vendor* show the result on the Devices page and in the Selected radio panel. A grey name with **?** is a weak guess: "some Apple device" says only who made it.

One device, many addresses. Phones, PCs and many smart-home devices change their Bluetooth address every few minutes. Sometimes the data still proves that addresses belong together: a Bluetooth Mesh device sends its own device number in every advert. Harava then puts them in one group. An Android phone or watch gives itself away in another way: it changes its address about every nine minutes, but the Google advertisement it sends stays the same over several changes, and one form of it carries four characters that stay for about a day. Harava joins the addresses that send the same advertisement, or the same four characters; in the panel the group reads *same Google advertisement* or *Google day code*. (Google does not document these advertisements. The day code is what four days of data showed: 270 codes, and never two different advertisements with one code at the same moment.) A service number that a maker uses in all its products is

not enough on its own (Ruuvi tags, Windows PCs and Galaxy Buds each share one), so such addresses are joined only when they also tell the same name — and only when the addresses are of the kind that changes. Twenty rental scooters that all call themselves "Scooter1" have twenty fixed addresses, and stay twenty scooters. With Group same device on (in Filter), a group is one row with an **N addr.** tag; its pin is the loudest address, and the hears are added up.

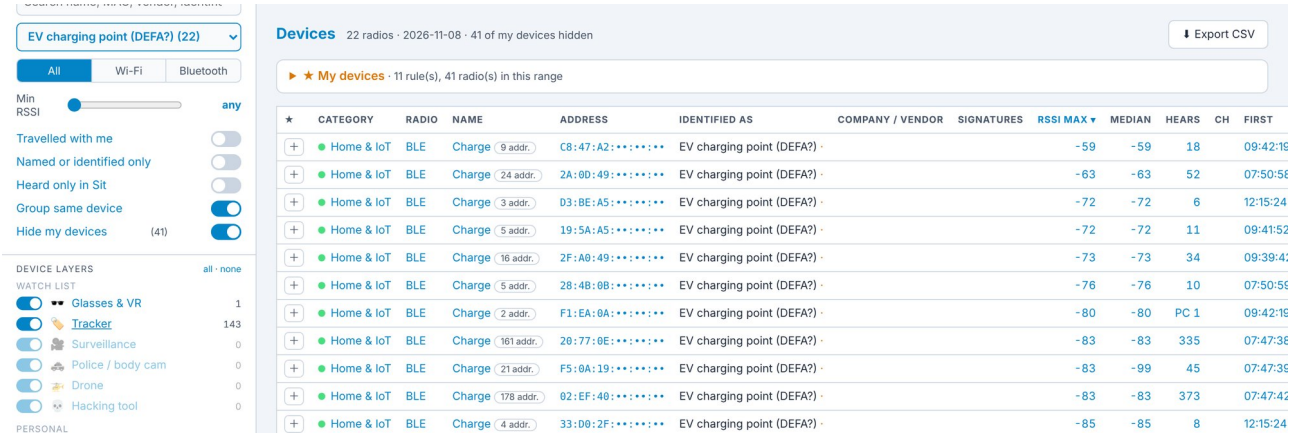


Figure 14. Identified as: EV charging point (DEFA?) on the Devices page. On 8 November, 957 addresses folded into 23 devices, most of them the charge points of the housing company's car park; one of them used 216 addresses during the day.

Charge points. The housing company's car park has a shared charging system of about 30 charge points. Its chargers advertise as Bluetooth Mesh devices named "Charge" that are not set up, and each one changes its address all the time. On 8 November Harava joined 937 such addresses into 22 charge points; most of them were heard only at 12:15, on the way to the car. A nameless address takes the name from the other adverts of the same device, so all its addresses are shown as a charge point. The maker (most likely DEFA) is not confirmed, hence the question mark.

Only one kind. The Identified as list under the search box shows only one kind of identified device — for example *Philips Hue lamp*, *Samsung TV* or *Oral-B electric toothbrush*. The list is grouped by device layer, so the *Home & IoT* group lists everything Harava found on that layer, each with its count after the other filters. Search finds the same texts too: `hue` finds the Hue lamp.

Making it better. The page lists the most common patterns that are still unknown — the best place to look for the next rule. Edit `ids/harava_ids.json` (it starts with a description of every field) and press Reload database; Harava does not need a restart. An identification is a pattern match like a signature: a strong hint, not proof.

13 My devices



Figure 15. My devices. A rule “Name is OTSOPC” hides the 31 addresses the author’s PC used during the day. The form at the top was filled by the + button of the D-Link router’s row; with Mask addresses on, Wi-Fi names show as *Wi-Fi network 1, 2* . . .

Your own router, PC, phone, earbuds, sensors and car are heard all the time and fill the lists. Mark them once and they disappear from the Devices list, the map and the Phone check while Hide my devices is on. Switch it off to see them again, marked ★.

Adding. Press + on a Devices row or + Mark as my device in the Selected radio panel: the form at the top of the Devices page is filled in. Check the label and press Add. You can also type a rule yourself.

Matching. *Address is* (one exact address), *Address starts with* (e.g. a router’s 2.4 and 5 GHz networks, whose addresses differ only at the end), *Name is*, *Name contains*; optionally only Wi-Fi or only Bluetooth. Bluetooth phones, PCs and earbuds change their address, so match them **by name**; + chooses that for them by itself. The column *Matches now* counts the radios a rule catches in the chosen range. Rules apply to every day, also to files imported later.

14 The Phone check

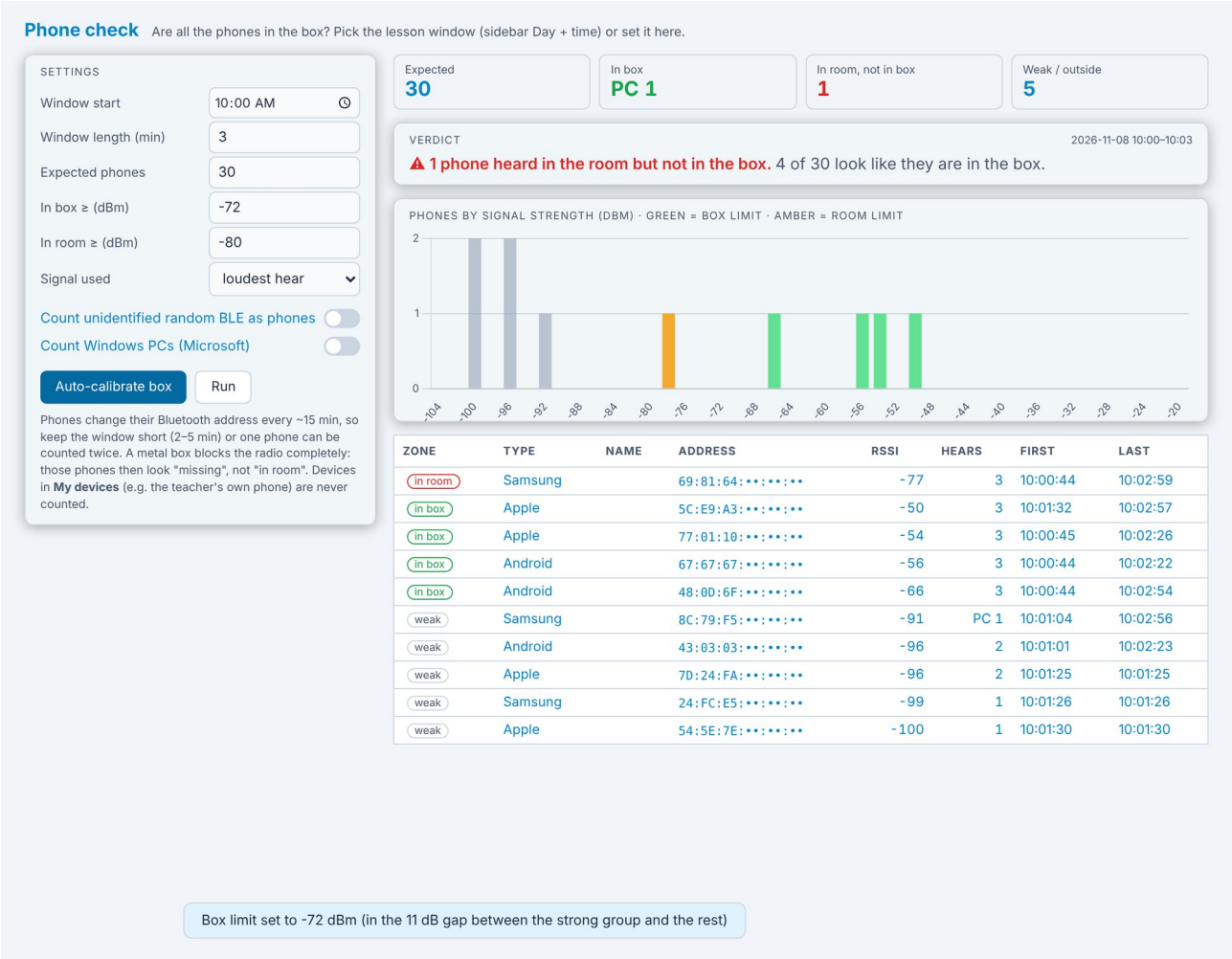


Figure 16. The Phone check over three minutes of the author's home log (not a classroom), after Auto-calibrate box with 30 phones expected: the strong group is "in the box", one phone is "in the room", the rest are weak.

The idea. At the start of a lesson the pupils put their phones into a box next to the teacher's phone, which runs Fieldwatch. The phones in the box are very close to it, so their signals are *strong* and similar. A phone that stays in a pocket or a bag in the same room is heard *more weakly*; phones outside the room are weaker still. The Phone check sorts the phones of a short time window into these three groups.

Step by step.

1. During the lesson, let Fieldwatch run with the log on. Afterwards export the log and import it.
2. Choose the day in the menu. In the Phone check, set Window start to a moment when the phones are in the box, and Window length to 2–5 minutes.
3. Set Expected phones, e.g. 30, and press Auto-calibrate box. If at least that many phones are heard, the box limit goes just below the 30th strongest; if fewer, it goes into a clear gap (at least 6 dB) between the strong group and the rest. It never goes below the room limit. You can type the limits yourself (defaults: box ≥ -55 dBm, room ≥ -80 dBm).
4. Read the cards and the verdict: *in box*, *in room, not in box*, *weak / outside*. The chart shows the phones by signal strength (green = box, amber = room). The table lists the phones, the ones in the room first.

What counts as a phone. Bluetooth radios of Apple, Samsung and Google / Android, without watches, earbuds, TVs and trackers. Two switches add Windows PCs and unidentified random addresses. iPads and Macs advertise like iPhones and are counted too. Your own devices (My devices) never count, so mark the teacher's phone and laptop as yours.

Things to know. Short window: phones change their address about every 15 minutes, so a long window can count one phone twice. **Silent phones**: a phone that is off, in flight mode or with Bluetooth off sends nothing and cannot be heard; a **metal box** blocks the signal too — those phones look *missing*, not *in room*. Use a cardboard or plastic box. **Calibrate once in the real room**, with a few known phones in the box and a few on desks: bodies, desks and walls weaken the signal. **The log only**: the Phone check does not use Sit exports, because a Sit row spans the whole session. **Privacy**: the check counts radios; it does not identify pupils. Agree its use with the school, keep the logs only as long as needed, and follow the school's rules and data-protection law.

15 Mask addresses

A Wi-Fi network's name together with its address can be looked up in public Wi-Fi maps, and your own home network points to your home. Before you share a screenshot, switch on Mask addresses under Map layers. It changes only what is drawn on the screen:

- every address keeps only its maker part: `40:86:CB:••:••:••`;
- every Wi-Fi network name — your own too — becomes *Wi-Fi network 1, 2 . . .* ;
- other people's PC names become *PC 1, 2 . . .* , and personal names such as *Anna's iPhone* become *•••'s iPhone*. Your own Bluetooth devices keep their names.

The data, the CSV export and the database are not changed, and Add in My devices still saves the full address. Switch it off to see everything again. All screenshots in this guide were taken with it.

16 Reading the data

A pin is where you were. Fieldwatch knows where *your phone* was, not where the radio was. The pin at the loudest hear is close to the radio only if you passed close to it. A radio heard from your sofa at −90 dBm may be in the next house.

Hears are not minutes. The log writes a radio the first time and then at every 25th hear; it forgets a radio after 15 minutes of silence (3 minutes for anonymous ones) and writes it again as new. A radio with 40 hears was not necessarily around longer than one with 4.

One gadget, many addresses. Random Bluetooth addresses change every few minutes; a phone nearby all day can be dozens of "radios". Match your own by name (My devices). Wi-Fi access points keep their address.

Travelled. Harava cuts your phone's own track into *stops* (the phone stayed within 150 m for at least five minutes) and movement, and then asks of every device where it was heard. There are two answers worth a tag:

- **at N stops** (amber) — heard at two or more of your stops that are at least 500 m apart. It was with you at both: it went with you, or it followed you. This is the strong sign. The panel lists the stops by their times.
- **same road** (grey) — heard for at least two minutes while you moved at least a kilometre, but never at two stops. In traffic this is normal: the car in front, a bus on the same motorway. It becomes interesting only if the same device shows up on another trip.

A radio you only passed gets no tag, however far apart the two sightings were: at 100 km/h half a kilometre takes 18 seconds. The addresses of one device are judged together, so a phone or a watch that changed its address five times on the way is one device that travelled, not five that did not.

On 8 November two radios were *at stops*: a D-Link 4G router (about −70 dBm at home, −29 dBm at the first stop 3.7 km away; it rode in the car that day) and a device named EF-R3PB2139 (the name fits an EcoFlow River 3 Plus power station). On a 23 km drive on 11 November the old rule (any address heard 500 m apart) marked 57 addresses; the levels show one device at three stops — almost certainly the driver's own watch, under 38 addresses — and a dozen devices on the same road for two to fourteen minutes: other cars.

Trackers. An AirTag or other tag *near its owner* is normal: every Apple user carries some. A tag *separated* from its owner and heard in several places along your way, over a long time, deserves a closer look. Fieldwatch and Harava read the mode from the raw data: for Apple Find My a full-key frame means separated; for Google Find Hub and DULT a flag byte tells it. Signals from one tag change address too, so look at the pattern, not at one address.

An "AirTag" is usually an iPhone. Every device of Apple's Find My network sends the same kind of message: AirTags, but also iPhones, iPads and Macs (so that they can be found), AirPods, and other makers' Find My items. Fieldwatch's *Apple AirTags* signature matches the message itself, so most of its "AirTags" are phones and computers — in the author's data of 8–10 November, 320 of 352. One byte of the message tells the kind of device, and Harava reads it:

- **Apple device (Find My)** and **Apple device, offline (Find My)** — an iPhone, iPad or Mac; moved to the *Apple device* layer. *Offline* is the full-key message of a device without a network connection (asleep, switched off or lost);
- **AirPods or Beats (Find My)** — moved to the *Audio* layer;
- **Apple AirTag** and **Find My accessory or newer AirTag** — stay on the *Tracker* layer. An accessory is another maker's item with Find My (key finder, bag, bike) or a 2nd-generation AirTag; the message alone cannot tell which;
- **Apple Find My device, kind not known** — a radio that only a Sit export has: a Sit carries no advertisement, so the byte cannot be read. It stays on the Tracker layer, but most such radios are iPhones too. Run the log together with the Sit.

The radio's panel still shows Fieldwatch's signature, with *layer by Harava* after it. The Find My message of a phone comes from another address than the phone's other messages, and the address changes often: the count of addresses is not a count of phones.

Signatures are patterns. A signature is a hint, not proof: an OUI (maker) match, a name or a data pattern. *Ray-Ban / Meta glasses* also matches Quest headsets; an *Apple Device* may be an iPhone, a Mac or an iPad.

Ruuvi sensors. Harava decodes Ruuvi tags (format 5): temperature, humidity, pressure and battery voltage, shown in the detail panel and as a tag in the list. On 8 November one read about −17 °C (a freezer) with the weakest battery, 2.55 V.

17 Questions you can answer

- *Were there smart glasses or VR headsets around me today, and when?* — Glasses & VR only, and Show: Glasses & VR only in the timeline.
- *Did anything travel with me that is not mine?* — Travelled with me on, Hide my devices on.
- *Is there a separated tracker along my way?* — Tracker only; look for pulsing red pins and *separated* tags.
- *What did the log miss?* — import the Sit export too and switch on Heard only in Sit.
- *Which cars, cameras or locks did I pass?* — the Vehicle, Camera and Smart lock layers.
- *Are all phones in the box?* — the Phone check (Section 14).
- *How cold is the freezer?* — search `Ruuvi` and open the tag.
- *What was around between 12:30 and 12:45?* — From–to times in the menu, or a click on the timeline.

18 Limitations

- Only what Fieldwatch heard: radios that were silent, too far, or behind metal are not there. A phone hears Bluetooth up to some tens of metres and Wi-Fi a little farther.
- Positions are the phone's positions; GPS indoors can drift by tens of metres, and the first minutes may have no position at all.
- The log is thinned (16 rows per scan, every 25th hear); without a Sit, brief radios can be missing.
- With the screen off, many phones park the Bluetooth scan; those stretches are marked on the timeline, and Bluetooth radios there were not heard at all.
- Signatures and identifications can be wrong in both directions, and one gadget can look like many; Harava groups addresses only when the data proves they belong together.
- A Sit export has no hear-by-hear times: Harava draws the timeline and the path from the log.
- The Phone check is an indicator for a teacher, not proof about a pupil.

19 Quick reference

I want to …	Do this
start Harava	double-click `start_harava.bat`; open `http://127.0.0.1:5057`
collect without gaps	Fieldwatch: Write to disk on + Start sit; export Log CSV and Sit CSV
bring in new files	copy them into `data`, Imports → Scan folder
see one day / an hour	Day, From–to, or click a timeline bar
see several days	Day and Until, or click a bar in Imports → Coverage
see when Bluetooth was paused	the amber bands on the timeline and ▨ Bluetooth paused in its title
see someone else's data	Whose data (shared imports; admins: All users)
see only one kind of device	click its name under Device layers (again for all)
graph only glasses	timeline Show → Glasses & VR only
look at one radio	click its pin or its row; × closes the panel
hide my own devices	+ on the row → Add; Hide my devices on
find what followed me	Travelled with me on
find what the log missed	import the Sit export, Heard only in Sit on
see what Harava identified	Identification; edit `ids/harava_ids.json`, Reload database
see one row per device	Group same device on
see only Hue lamps	Identified as → Philips Hue lamp
share a screenshot	Mask addresses on (Map layers)
change my password	click your name in the menu (*My account*)
check the phone box	Phone check: window start, 3 min, expected, Auto-calibrate box
take the list to Excel	Devices → Export CSV
remove a wrong file	Imports: switch off or Delete
share an import	Imports → Share
see what is new	What's new under the name
remove the map watermark	CARTO key in `api_keys.py`
start from scratch	stop Harava, delete `harava.db`

Table 7. Quick reference.

Harava reads the logs of Fieldwatch by Off Grid Pete (`https://github.com/OffGridPete/Fieldwatch`); its device layers come from the Fieldwatch signature catalog (MIT licence). Harava itself is not part of Fieldwatch. Leaflet (BSD-2) and Chart.js (MIT) are bundled; the map background is © OpenStreetMap contributors © CARTO.